

UniTS Anomaly Ratio Sweep

STPSat4-ADCS | Generated 2026-03-12 10:53

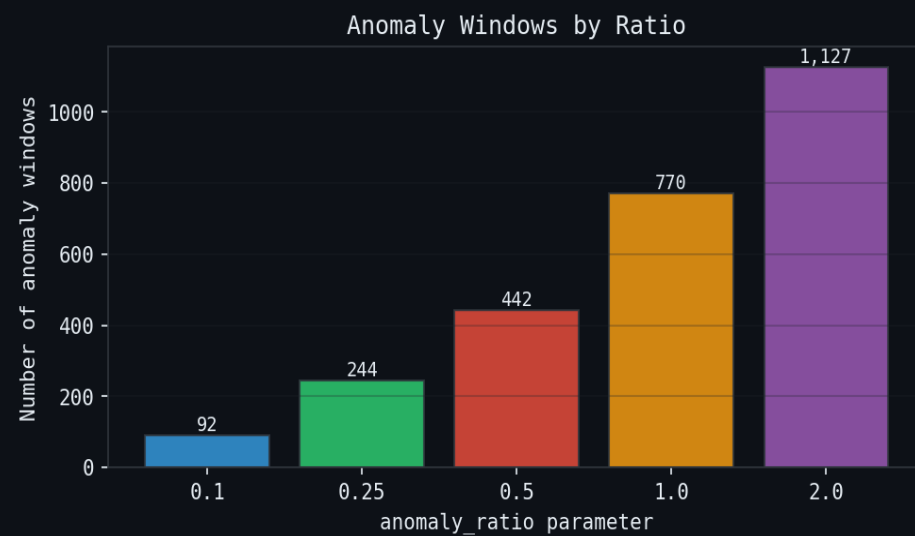
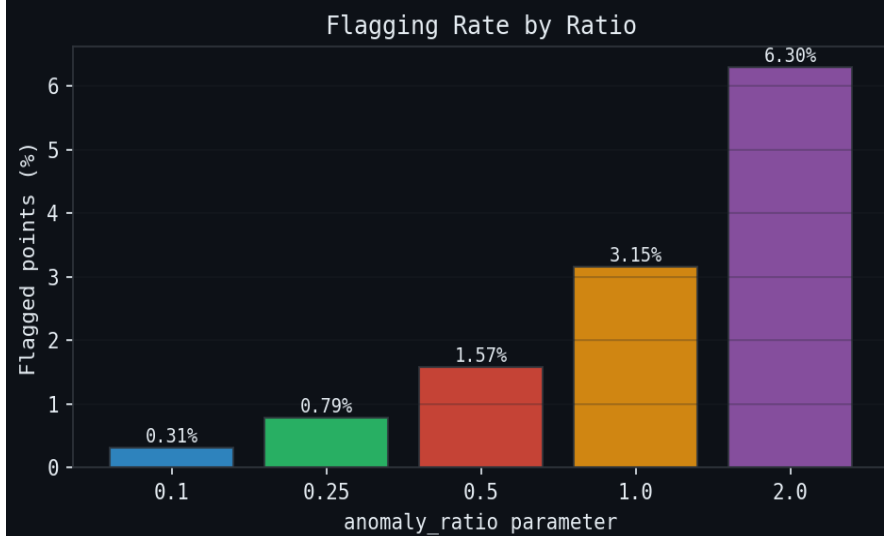
Results by Ratio

anomaly_ratio	Threshold	Flagged Points	Flag Rate	Windows
0.1	3.296293	137	0.314%	92
0.25	1.869381	343	0.787%	244
0.5	1.142016	686	1.574%	442
1.0	0.714291	1,373	3.150%	770
2.0	0.436029	2,746	6.300%	1,127

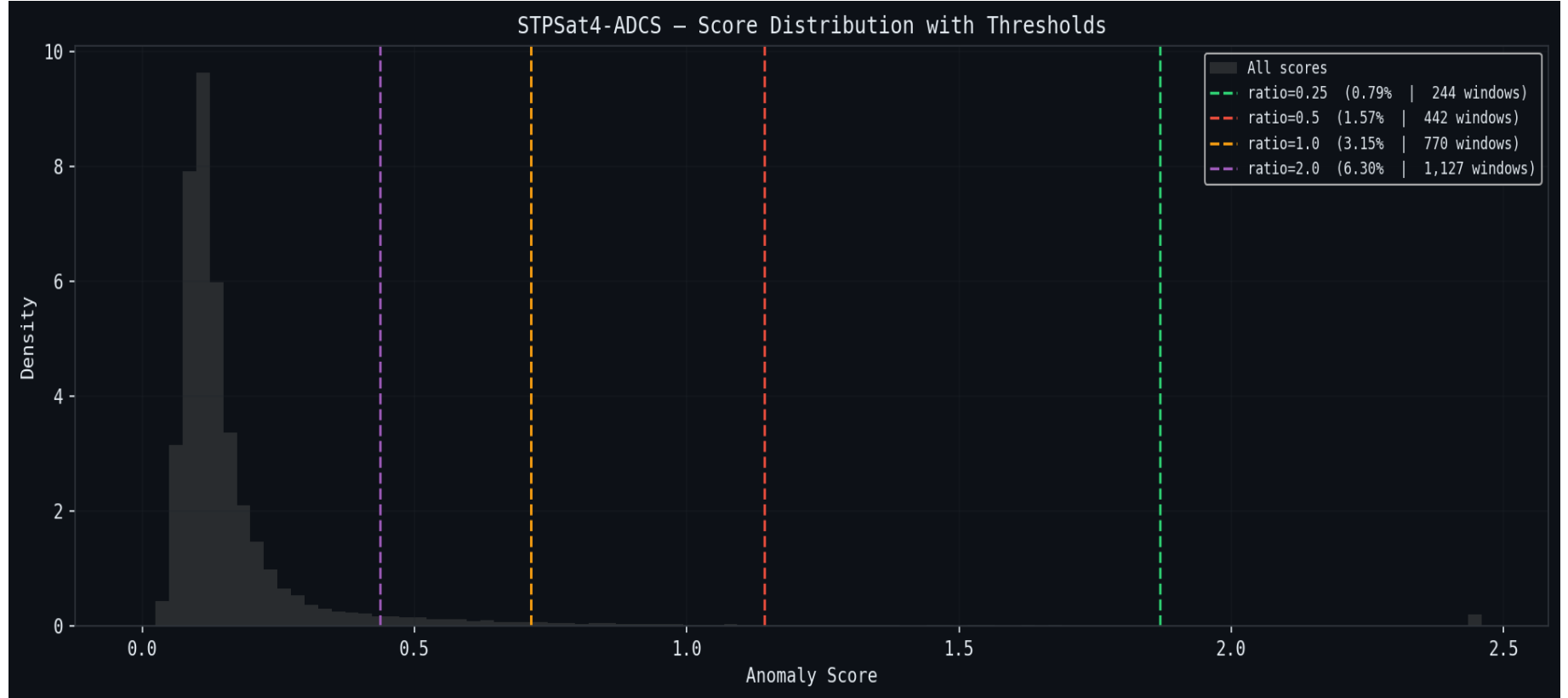
anomaly_ratio controls the detection threshold: UniTS flags the top ratio% of reconstruction error scores as anomalies. Lower values = stricter threshold = fewer flags. The threshold column shows the exact score cutoff for each ratio.

Flagging Rate & Window Count

STPSat4-ADCS – Ratio Sweep Summary



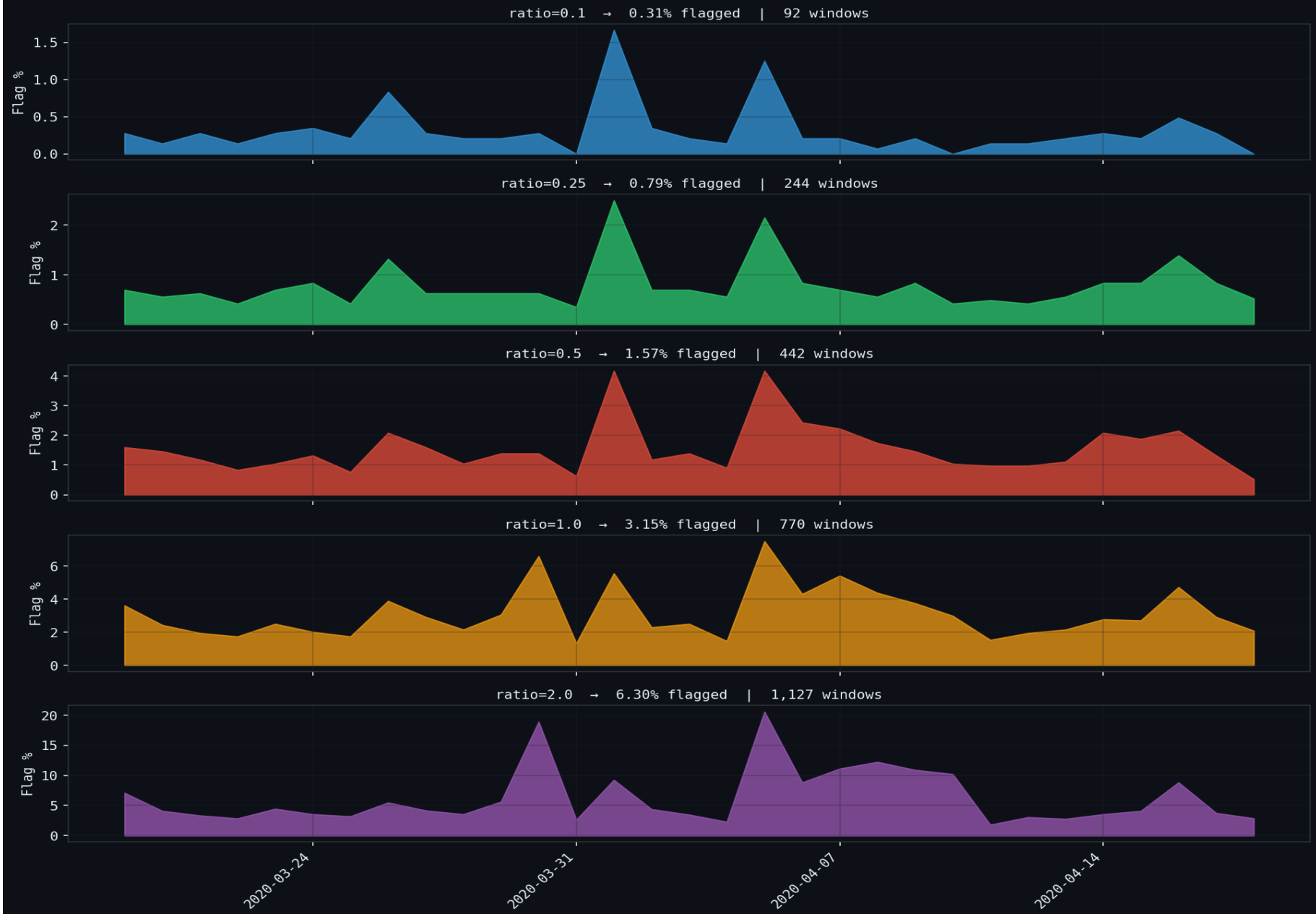
Score Distribution with Thresholds



Dashed vertical lines show where each ratio sets its threshold. Choose a ratio whose threshold sits clearly in the tail of the distribution rather than in the main body of normal scores.

Daily Flagging Rate by Ratio

STPSat4-ADCS – Daily Flagging Rate by Ratio



A realistic anomaly rate for a nominal satellite is typically 0.1–1%. Look for a ratio that shows isolated spikes rather than sustained elevated flagging.